

Purple Team Outline

Week 02: Network Discovery & Reconnaissance

Objective: To contrast aggressive scanning techniques with stealthy discovery and implement the first layer of defensive detection.

Task 01: Active & Passive Discovery

- **Description:** Map the network using both "loud" and "silent" methodologies.
- **Technical Requirements:**
 - Perform Active Discovery using ARP/ICMP sweeps to identify live hosts
 - Execute a TCP SYN Scan (Stealth Scan) and compare its fingerprint to a Full Connect Scan.
 - Conduct Service Identification to determine versions of running applications (Fingerprinting).
- **Learning Goal:** Understand the trade-off between speed of discovery and the likelihood of being detected by a SOC.
- **What to Document :**
 - Make a difference between **Reconnaissance** Methods (**Active & Passive**) & Define with a valid reason why Reconnaissance is important when you are pentesting ?
 - Perform **Dns** (AAAA, CNAME, MX) & **Whois** enumeration on any legal website (example.com) -- Attach the results screenshot.
 - Perform an Stealthy Scan by nmap & capture the **TCP SYN SCAN** with **wireshark** – Attach the captured packet's screenshot.
 - Define in your own words which scanning technique is most commonly used when you do not want yourself to be detected (TCP Full Scan, TCP SYN Scan).
 - Identify the versions running on services with **nmap/zenmap** – Attach the version screenshots.

Task 02: Defensive Summary & IDS Implementation

- **Description:** Analyze the "Attacker" footprints from a defensive perspective.
- **Technical Requirements:**
 - Use Wireshark to analyze the specific packet patterns generated by Nmap's stealth scans.
 - Implement a basic IDS/Defensive Summary to alert on high-frequency connection attempts.
- **Learning Goal:** Learn how signature-based systems catch scanners by identifying non-standard packet flags.
- **What to Document :**
 - Installation of Snort & What version you are using of snort -- Attach the Screenshots.
 - Create an alert for (ICMP) to allow only a particular number of packets to send – Attach the **Local.rules** file screenshot.
 - How is ICMP Different from ARP ? In your own words.
 - Start the Snort & start sending the packets **ICMP** from your Attacking machine to the victim machine – Attach the screenshots.
 - In the **/var/snort** directory open the captured logs of alert you created – Attach the log screenshot.
 - What do you understand about the ICMP flooding attack & Smurf attack in context of **Dos** & **DDos**?